

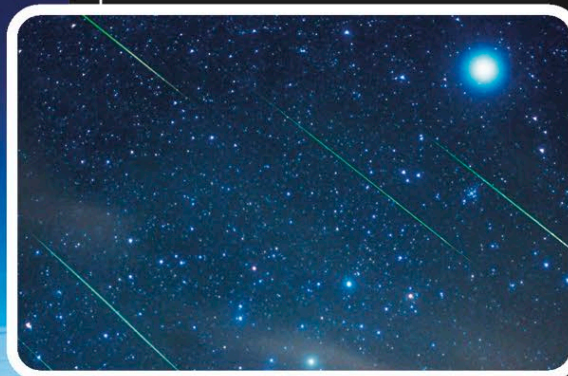
Protecting the crew

The Crew Module of the Orion spacecraft has a heat shield made of a material called AVCOAT, which disintegrates into hot plasma as it enters Earth's atmosphere. Underneath lies a carbon-fiber dome and a strong titanium structure that supports the body of the craft.

RETURNING TO EARTH

HEAT SHIELDS

Spacecraft reentering the atmosphere travel at incredibly high speeds. It takes energy to push through the air, and this is converted into heat. The temperature of the spacecraft can rise to as high as 5,000°F (2,800°C). To protect the astronauts inside, the spacecraft is designed with a special heat shield that disintegrates to form a hot gas called plasma, which carries away heat. The plasma is so hot that it glows with a bright yellow light.

**SHOOTING STARS**

Meteors, or shooting stars, appear when chunks of rock from space enter Earth's atmosphere at high speed. Their passage through the air generates heat, which disintegrates the rocks, creating a streak of glowing plasma across the sky.

The Orion crew module can carry 4 astronauts and everything they need to survive for up to 21 days.

PLASMA

Plasma is made of gas atoms (see page 177) that are so energized that they lose their negatively charged electrons. They then form particles called ions, which have a positive charge. Plasma is found throughout the Universe, such as in flames, lightning bolts, and the surface of the Sun.

